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Evaluation of different materials used for sealing of implant abutment access channel and the peri-implant sulcus microbiota: A 6-month, randomized controlled trial

Caroline V. Rubino[1] | Brandon G. Katz[1] | Kylie Langlois[2] | Howard H. Wang[1] | Julio A. Carrion[1] | Stephen G. Walker[3] | Jackie L. Collier[2] | Vincent J. Iacono[1] | Srinivas R. Myneni[1]

[1]Department of Periodontology, School of Dental Medicine, Stony Brook University, Stony Brook, NY, USA

[2]School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, NY, USA

[3]Department of Oral Biology and Pathology, School of Dental Medicine, Stony Brook University, Stony Brook, NY, USA

Correspondence
Srinivas R. Myneni, Department of Periodontology, Stony Brook School of Dental Medicine, South Drive, Stony Brook, New York, NY 11794, USA.
Email: srinivas.mynenivenkatasatya@stonybrookmedicine.edu

Jackie L. Collier, School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, New York, NY 11794, USA.
Email: jackie.collier@stonybrook.edu

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Abstract

Objective: Peri-implantitis has been attributed to a myriad of factors, including microleakage at the abutment-implant interface. Implant abutment access channel sealing materials (IACSM) are readily used in implant dentistry, with little evidence on their effect on microleakage. This study aims to evaluate the effect of IACSM on the microbial composition in the implant access channel and the peri-implant sulcus.

Methods: A total of $n = 8$ patients (64 implants) were included in this single-blinded, randomized controlled trial, whereas four different materials (cotton, polytetrafluoroethylene [PTFE], synthetic foam, or polyvinyl siloxane [PVS]) were randomly placed as an IACSM. Following 6 months, microbial analysis was completed on the IACSM and samples from the peri-implant sulci via PCR and high-throughput sequencing. Bacterial samples on the IACSM and in the peri-implant sulci were classified according to Socransky's microbial complexes.

Results: There was a preponderance of early colonizing bacteria within the IACSM, while the peri-implant sulci were dominated by Orange complex bacteria. The proportion of Red and Orange complex members on the IACSM was significantly less than in the peri-implant sulci. The proportion of Green, Yellow, and Blue complex members found on the IACSM was significantly greater than in the peri-implant sulci. *Atopobium*, a diverse species not included in the microbial complexes, was frequently detected in the peri-implant sulcus samples.

Conclusions: No detectable effects of IACSM on the microbial community in the peri-implant sulcus or on the IACSM were identified. Variation of bacterial species was most dependent on the individual patient. No significant differences were found in the periodontal parameters between the different treatment groups.

KEYWORDS

abutment access channel sealing materials, dental implants, dental prosthesis, fixed dental hybrid prostheses, oral microbiome, peri-implant microbiota, peri-implant sulcus

This present study was conducted from 2013–2016 at Stony Brook University, Stony Brook, New York.

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1 | INTRODUCTION

Studies of peri-implant diseases have elucidated many contributing factors that predispose an individual to developing peri-implant mucositis or peri-implantitis. Peri-implant mucositis parallels gingivitis, as it presents with gingival inflammation, suppuration, and is often associated with probing depths exceeding 4 mm in the absence of alveolar bone loss around an implant (Heitz-Mayfield & Salvi, 2018). Unlike peri-implant mucositis, peri-implantitis is characterized by progressive crestal bone loss, signs of inflammation, and probing depths >6 mm (Schwarz et al., 2018). History of periodontitis, lack of an implant maintenance regimen, diabetes, and smoking are patient-dependent risk factors for peri-implant diseases (Roos-Jansåker et al., 2006). Other contributing factors include the location of the implant in the alveolar arches, types, and dimensions of implants, cemented versus screw-retained restorations, number and prosthetic positioning of implants, length of time of implant placement to restoration, and loading protocols (Daubert et al., 2015; Ferreira et al., 2015; Marcantonio et al., 2015; Poli et al., 2016).

The etiology and pathogenesis of peri-implantitis are analogous to that of periodontitis (Meffert, 1996). Similar to periodontal pathogens around natural teeth, the peri-implant microbiome changes with increasing probing depths. Healthy implants with shallow probing depths are primarily colonized with Gram-positive facultative bacteria such as *Streptococcus* and *Actinomyces* (Mombelli & Décaillet, 2011; Mombelli & Mericske-Stern, 1990). Peri-implantitis sites demonstrate an increase of Gram-negative species such as *Fusobacterium nucleatum*, *Porphyromonas gingivalis*, *Tannerella forsythia*, *Treponema denticola*, and *Prevotella intermedia* (Albertini et al., 2015; Sbordone et al., 1995). Multiple studies, including a 3-year longitudinal study, demonstrated that gingival inflammation and increased probing depths in the peri-implant tissues are significantly associated with increased biofilm formation and numbers of motile rods and spirochetes (Adell et al., 1986; Pontoriero et al., 1994). This shift in bacterial species to a more Gram-negative flora during the development of peri-implantitis may be driven by microleakage at the abutment-implant interface (A-I). However, the clinical significance has not been well studied (Quirynen et al., 1994).

Microleakage of fluids containing bacterial byproducts and nutrients required for bacterial growth at the abutment-implant (A-I) interface might be a contributing factor in the development of peri-implant diseases. (Gross et al., 1999). Randomized control clinical trials have demonstrated no significant microbiological differences between tapered interference fit and taper-integrated screwed-in types of implant abutment connections, whereas systematic reviews tend to indicate that all the connections presented an amount of microgap and bacterial microleakage, albeit conical and mixed connection systems deliver better outcomes (Öztürk et al., 2017; Lauritano et al., 2020, Berberi et al., 2014).

In previous in vitro studies, bacteria were shown to migrate into implant abutment assemblies immersed in blood serum inoculated with bacteria (Piattelli et al., 2001). These authors hypothesized that the microgap and associated microleakage create a zone of inflammation, which contributes to the incidence of peri-implantitis. The zone of inflammation is characterized by an inflammatory cell infiltrate at the junction of the connective tissue and the implant abutment interface, irrespective of whether the abutments were kept devoid of plaque or were allowed to accumulate plaque. This zone of inflammation is likely attributed to the presence of microorganisms within the microgap (Ericsson et al., 1995). Endotoxin microleakage occurring at the A-I interface after experimental inoculation inside the implant body has been shown to induce an upregulation of pro-inflammatory mediators and cytokines such as Toll-like receptor 9 (TLR), tumor necrosis factor-α (TNFα), interleukin 1β (IL-1β), interferon 6 (IL-6), interferon gamma (IFN-γ), and nuclear factor kappa beta (NF-κβ), which may lead to alveolar bone loss and the clinical manifestations of peri-implantitis through pathologic osteoclast activation (Harder et al., 2012).

A myriad of factors has been found to affect the degree of microleakage at the A-I interface. Cement-retained implants tend to have lower bacterial load at the inner portions of the A-I interface, but significantly higher bacterial load at the peri-implant sulci when compared to screw-retained implants (Penarrocha-Oltra et al., 2016). Implants with an external connection have been correlated with higher levels of microleakage when compared to implants with internal or Morse taper connection (Larrucea Verdugo et al., 2014).

Current research describes peri-implantitis as a microbiologically heterogeneous infection with predominantly Gram-negative species and the in vitro studies provide evidence that the A-I interface can be a nidus of infection, harboring and exchanging bacteria and/or bacterial endotoxins that cause pronounced inflammation (Do Nascimento et al., 2009; Quirynen et al., 1994). While different strategies such as modifying implant connection designs and prosthetic screw torque values are being explored in reducing this source of bacterial leakage, another readily modifiable parameter that requires additional investigation is the implant prosthesis access channel sealing materials (IACSM). Employing different IACSM may decrease bacterial contamination at the A-I interface. To the authors' best knowledge, there is currently only one in vitro study testing the effect of different IACSM on bacterial contamination. The investigators concluded that when sealing the access holes of screw-retained implant prostheses, gutta-percha or polyvinyl siloxane would help reduce microleakage (Park et al., 2012). To date, there are no clinical data evaluating effects of IACSM on implant abutment access channel and sulcus microbiomes (Park et al., 2012).

The aim of this study was to evaluate the effect of different IACSM on bacterial species found in both the implant abutment screw channel and the peri-implant sulcus in an in vivo 6-month randomized controlled trial. In addition, periodontal parameters such as plaque index, gingival index, and probing depths were assessed.

2 | MATERIAL AND METHODS

A single-blind, randomized controlled trial of 6-month duration was conducted to examine the correlation between the bacterial DNA

found on four different IACSMs (cotton, polytetrafluoroethylene (PTFE), synthetic foam, polyvinyl siloxane (PVS), and the bacterial DNA) found in the peri-implant sulci. Ethics committee approval was gained from the Institutional Review Board at Stony Brook University (IRB# 409069). All procedures conducted in this present study were in accordance with the 1964 Helsinki Declaration. The present study was conducted from 2013–2015 and has been prepared in accordance with guidelines outlined in the CONSORT statement for reporting of randomized controlled trials (Schulz et al., 2010).

2.1 | Inclusion and Exclusion criteria

Inclusion criteria were as follows: (1) patients between the ages of 18–85 years old; (2) edentulous patients treated with full arch, fixed implant-supported prostheses.

Exclusion criteria were as follows: (1) poor oral hygiene and motivation; (2) uncontrolled diabetes; (3) pregnant or lactating; (4) drug abusers; (5) current smokers; (6) psychiatric problems or unrealistic expectations; (7) acute infection around implant of interest; (8) positive to HIV, or hepatitis B, or hepatitis C; (9) affected by other autoimmune diseases; (10) under chronic treatment with steroids or non-steroidal anti-inflammatory drugs.

Subjects were identified from a patient electronic database and contacted by telephone to determine their interest in participation in the study. A total of eight fully edentulous subjects with both maxillary and mandibular arches rehabilitated with full arch fixed implant-supported prostheses were included in the study.

2.2 | Clinical protocol

The study consisted of two or three subject interactions over a period of 1 year from recruitment. Initial visit consisted of screening of the patient and consent for participation in treatment. Visit two (T_1) consisted of removal of the prosthesis, followed by debridement of the prosthesis and implants. Hybrid prostheses were placed in an ultrasonic bath, followed by manual debridement with a denture brush and then soaked in 0.12% chlorhexidine gluconate oral rinse (Peridex™) before replacement. Implants were debrided via hand scaling with titanium curettes, followed by thorough irrigation of the peri-implant sulcus and the implant abutment access channel with 0.12% chlorhexidine gluconate oral rinse (Peridex™) prior to replacement of the prosthesis. The prosthesis was then replaced, followed by placement of the IACSM based on a predetermined randomization scheme. Four different materials were utilized as IACSM; cotton, (Bleached Cotton; Richmond Dental), PTFE (Merco M55 Thread Teflon Seal Tape; Merco Co.), synthetic foam (EndoRing® E-Foam® Inserts; Jordco®), and PVS (Aquasil; Dentsply). Materials were evenly distributed per patient ($n = 2$ implants per material type, $n = 8$ implants per subject) and randomly placed as predetermined by a statistician. Once the IACSM was placed, the prosthesis was replaced, all prosthetic screws were hand tightened by a single provider, and a composite resin was used to seal over the IACSM and implant abutment access channel (Filtex™ Supreme Ultra Universal Restorative; 3 M™ ESPE™). In some cases, Visit one and two were combined. Oral hygiene was reviewed with all participants at Visit 2 (T_1). Visit three (T_2) consisted of IACSM retrieval at approximately 6 months and collection of the sulcus biofilm by placing 2 sterile paper points in the deepest site of the peri-implant sulcus for 10 s and transferring them into sterile Eppendorf tubes. The samples were stored at −80°C until processed for DNA extraction. In addition, at both T_1 and T_2, clinical data (probing depth, plaque index, gingival index) were collected while the prosthesis was removed. Upon replacement of the prosthesis, prosthetic screws were hand tightened as per manufacturer's recommendations and completed by a single provider for all subjects at T_1 and T_2. Probing depth (PD) was defined as the distance from the mucosal margin to the base of the peri-implant sulcus. Silness and Loe Plaque Index (PI) and Loe and Silness Gingival Index (GI) were utilized (Löe, 1967).

2.3 | Evaluation of dental materials

The dental materials (cotton, PTFE, synthetic foam, PVS, Paper Point) were tested for the presence and growth of microbes prior to use intraorally, to rule out contamination of the materials prior to placement in the abutment channels. The materials were placed aseptically into sterile collection tubes and then nutrient broth was added. The tubes were allowed to incubate at room temperature for 4 days. The tubes were then incubated at 37°C for another 4 days. Following this period, 1 ml of medium was removed from each tube and was concentrated via centrifugation, while the remainder was incubated for another 20 days at room temperature.

2.4 | DNA extraction

DNA was extracted using MO BIO Powersoil kit (Qiagen). This kit was selected to isolate the DNA from our samples because it is intended for use with environmental samples, including difficult to process soil types. As this was a novel study, we were unsure what, if any, DNA could be isolated from the different materials used in this study. This kit has been successfully used for isolation of PCR-amplifiable DNA from diverse sample types and organisms, including bacteria, algae, and actinomycetes. The manufacturer's protocol was followed with the following modifications: The dental materials and paper points were individually transferred to the PowerBead Tubes. The Eppendorf tubes in which the materials had been stored were washed with the first solution in the protocol and the wash was transferred to the appropriate PowerBead Tube. This was an attempt to recover any cells which may have been contained in fluid collected with the samples that did not get transferred with the bulk dental material to the PowerBead Tube. Control samples, which were sterilized samples of each dental material and paper points,

were also transferred in the same way. Care was taken to use similar volumes of these control materials to the volumes of the test materials. Extracted DNA was quantified by PicoGreen (Invitrogen) and then stored at −20°C until used as a template for PCR.

2.5 | PCR, sequencing, and analysis

Microbial community structure was determined by sequencing 16S small subunit ribosomal RNA (rRNA) using primers S-D-Arch-0519-a-S-15 (A519F; CAGCMGCCGCGGTAA) and S-D-Bact-0785-b-A-18 (802R; TACNVGGGTATCTAATCC) (Klindworth et al., 2013). The forward primer for each sample contained a unique barcode which allowed sequences to be assigned to the proper sample during data analysis. A 30 cycle polymerase chain reaction (PCR) using the HotStarTaq Plus Master Mix Kit (Qiagen) was run under the following conditions: 94°C for 3 min, followed by 28 cycles of 94°C for 30 s, 53°C for 40 s and 72°C for 1 min, after which a final elongation step at 72°C for 5 min was performed. After amplification, PCR products were assessed by electrophoresis in a 2% agarose gel to determine the success of amplification and the relative intensity of bands. Multiple samples were pooled together (e.g. 100 samples) in equal proportions based on their molecular weight and DNA concentrations. Pooled samples were purified using calibrated Ampure XP beads. Then, the pooled and purified PCR product was used to prepare a DNA library by following Illumina TruSeq DNA library preparation protocol. Sequencing was performed at MR DNA (www.mrdnalab.com, Shallowater, TX, USA) on a MiSeq following the manufacturer's guidelines. The sequence data were submitted to the NCBI Sequence Read Archive (https://www.ncbi.nlm.nih.gov/Traces/study/) under BioProject accession number PRJNA649980.

Sequence data were processed in two ways. The first used MR DNA's analysis pipeline (MR DNA). In summary, sequences were joined, assigned to samples, depleted of barcodes, then sequences <150 bp or with ambiguous base calls removed and remaining sequences were denoised. Operational taxonomic units (OTUs) were defined by clustering at 3% divergence (97% identity) followed by removal of singletons and chimeras (Capone et al., 2011; DeSantis et al., 2006; Dowd, Callaway, et al., 2008; Dowd et al., 2008; Edgar, 2010; Eren et al., 2011). This pipeline yielded 4,097,491 sequences, ranging from 10,344 to 247,318 per sample with a mean of 67,168. The final 492 OTUs were taxonomically classified using BLASTn against a curated database derived from GreenGenes, RDPII, and NCBI (http://rdp.cme.msu.edu, www.ncbi.nlm.nih.gov) and compiled at each taxonomic level into OTU abundance tables. The second processing pipeline was carried out in QIIME2 (v.2017–12), run through a 64-bit Virtual Box (VB) (v.5.2.4) (Caporaso et al., 2010). After the raw fastq files from MR DNA were demultiplexed in QIIME2, the R package dada2 was used to dereplicate and merge all sequences into an abundance table of 99% identity amplicon sequence variants (ASV) (Callahan et al., 2016). After rarefaction in QIIME2, there were 17,146 sequences per sample. One sample was lost in this step because too few good-quality sequences were obtained from it. ASV present in only one sample were removed, reducing the initial 1,219 ASV to 571 ASV and 1,206,863 sequences. The trimmed and rarefied ASV abundance table and representative set of sequences were analyzed through various QIIME2 commands, automated with C++ in the text editor Atom (v.1.27.0), using the compiler package atom-gpp (https://github.com/kriscross07/atom-gpp-compiler). All automated commands can be found in the following repository: https://github.com/kylielanglois/QIIME2_automated_commands. Briefly, the representative set of sequences was aligned and used to produce a phylogenetic tree, and principal coordinate analysis (PCoA) coordinates were produced using the weighted UniFrac distance metric (Lozupone & Knight, 2005). A PCoA ordination plot was produced in RStudio using the R package "ggplot2" to visualize the beta diversity within the dataset (Wickham, 2016). ASV were assigned taxonomic identifications from the Silva database (132 release, 16S only, 99% identity).

2.6 | Statistical analyses

A 4 × 4 Latin Square design was adopted to randomly assign four materials to eight implants per patient (Table S1). The design ensured each implant was evenly assigned to the four materials, while treating each patient as a random block. Due to lack of preliminary data, effect size was used to calculate the statistical power based on a *F* test of Repeated ANOVA. Assuming a two-sided alpha of 0.01 (more contingent than 0.05 for multiple comparisons) and a ratio of within-subject variance to the total variance at a conservative level of 33%, a twice-replicated 4 × 4 Latin Square design with 8 patients will provide 97% power to detect a significant difference of 1.2 standard deviation among materials. A total of sixty-four material samples, sixteen of each material, were obtained. One paper point was used per implant and pooled into one of four Eppendorf tubes per patient by type of IACSM. A total of 128 paper points were obtained. Thirty-two paper point samples, one of each per IACSM type within the same patient, were analyzed.

Comparisons of bacterial community structure among groups of samples were performed on both the OTU and ASV datasets using PERMANOVA (adonis function in the R package "vegan") (Oksanen J, 2013). To facilitate further statistical analysis and interpretation, the bacteria identified were grouped into the Socransky and Haffajee bacterial complexes and then noted as "other" when not falling into any of the complex categories (Socransky et al., 1998). T-tests were done to determine whether there was a difference between the clinical parameters (PD, PI, GI) taken at T_1 and T_2 or between the proportion of complex bacteria on the IACSM and the peri-implant sulcus. The correlation coefficient (R) was calculated to investigate the relationship between the proportion of each bacterial complex on the IACSM and the peri-implant sulcus. Pearson coefficients were calculated to test the correlation between Red complex bacteria and the probing depth at T_2, and the correlation between each microbial complex found in the peri-implant sulcus. ANOVA was performed to determine the effect of several factors on the microbial complexes. The first was the effects of the IACSM, position, and patient for each microbial complex found in the peri-implant sulcus. The second was

the effects of the IACSM, position, and patient for each microbial complex found in the IACSM as compared to the microbial complex found in the peri-implant sulcus. The third and final was the effects of the IACSM, position, and patient for each of the "other" bacteria found in the peri-implant sulcus.

3 | RESULTS

3.1 | Patient demographics

Six of the eight subjects fulfilled the criteria of eight implants under a maxillary prosthesis. One patient had only seven implants in the maxillary arch (Patient 6), and one had eight implants distributed under both maxillary and mandibular arches in an "All-on-Four" design for each arch (Patient 8). The study population comprised of 6 males and 2 females of Caucasian descent with an average age of 60 years old. The study prostheses were in function an average of 4.43 years, with the least time in function being 1.36 years (Patient 8) and the most time in function being 7.94 years (Patient 6). The study time between the first clinical session (T_1) and second clinical session (T_2) in this study averaged 6.8 months (*SD* = 29.87), ranging from 175 days (Patient 7) to 265 days (Patient 5) (Table 1). All implant abutment connections studied were flat, butt-joint connections.

3.2 | Clinical outcomes

Selected clinical data are summarized in Table S2. Among the tested specimens, the average probing depth at time one (PD at T_1) per site (as defined by material) ranged from 2.33 mm (Patient 5, teflon) to 9.33 mm (Patient 7, foam). The average PD at T_1 among patients was 3.87 mm (*SD* 1.58). 18.75% of sites were at or greater than 4 mm but less than 6 mm, 9.375% were at or greater than 6 mm but less than 8 mm; and 3.125% were at 8 mm or greater. The average probing depth at time two (PD at T_2) per site (as defined by material) ranged from 2.0 mm (Patient 5, teflon) to 9.17 mm (Patient 7, foam). The average PD at T_2 among patients was 3.96 mm (*SD* 1.69). It was found that 71.875% of PD across all sites were below 4 mm; 15.625% were at or greater than 4 mm but less than 6 mm, 6.25% were at or greater than 6 mm but less than 8 mm; and 6.25% were at 8 mm or greater. There was no statistical difference between the PD at T_1 and T_2, as determined by *t* test ($p = .3306$).

The average PI at T_1 among all patients was 1.89 (*SD* 0.56). A PI = 1 at T_1 was evident for 21.875% of all sites across all patients, while 65.625% had an index of 2, and 12.5% had an index greater than 2. The average PI at T_2 among all patients was 1.77 (*SD* 0.58). A PI = 1 at T_2 was evident for 31.25% of all sites across all patients, while 59.38% had an index of 2, and 3.38% had an index greater than 2. There was no statistical difference between the PI at T_1 and T_2, as determined by T-Test ($p = .211$) (Table 1). A GI = 1 at T_1 of 1 was evident for 12.5% of all sites across all patients, while 75% had an index of 2, and 12.5% had an index greater than 2. The GI at T_2 per site (as defined by material) ranged from 1 to 3. The average GI at T_2 among patients was 1.97 (*SD* 0.46). A GI = 1 at T_2 was evident for 12.5% of all sites across all patients, while 75% had an index of 2, and 12.5% had an index greater than 2. The measurements of GI at T_1 and T_2 for all sites and patients were identical, so no *p*-value was calculated.

TABLE 1 Summary of patient demographics

Age (Years)	Range	53–73
	Mean	60
	Standard deviation	6.52
Sex	Male	6
	Female	2
Race	White	5
	Black	2
	Hispanic	1
Time study prosthesis in function (Years)	Range	1.36–7.94
	Mean	4.43
	Standard deviation	1.97
Time between T_1 and T_2 (Days)	Range	175–265
	Mean	206.86 (about 6.8 months)
	Standard deviation	29.87
Implant type	Nobel replace (Trilobe connection)	48
	3i Osseotite Certain (Internal hexagon connection)	8
	Brånemark (External hexagon connection)	8

3.3 | Bacteria in materials and Peri-implant sulci

After incubation in nutrient medium for 28 days, dental materials not exposed to patients showed no growth of bacteria by turbidity or microscopy. No detectable DNA (by PicoGreen, which has a limit of detection of ~0.1 μ g DNA per μl) was extracted from the control materials, although they did produce PCR products and sequencing results (Figure 1). Most (19) of the 32 IACSM samples recovered from patients and 4 of the 32 sulcus samples (mainly from patients 2 and 3) also yielded no detectable DNA but also did produce PCR products and sequencing results.

In both the OTU and ASV 16S amplicon sequencing analyses, multivariate ordinations of the whole community suggested that the sulcus samples were distinct from the IACSM samples, for example grouping to the right and left (respectively) of Axis 1 (which captured 39% of the variance) in the ASV analysis (Figure 2). PERMANOVA confirmed that the bacterial communities of the sulcus and IACSM samples were significantly different from each other ($p < .001$), and that each sample type was significantly different from the sterile control materials ($p < .001$ in each comparison). PERMANOVA also showed that the bacterial communities of the sulcus samples were significantly different by patient ($p < .001$), but not by material. At the whole community level, IACSM samples were not significantly different by patient or by material.

The taxa most frequently detected were *Fusobacterium nucleatum* (~23% of all sequences), followed by *Atopobium* species (~4% of all sequences), *Streptococcus oralis*, and *Lactobacillus johnsonii* (each ~3% of all sequences). An average of 43.12% ($SD = 10.72$) of the DNA sequences representing the bacterial communities was from bacteria not classified into the microbial colored complexes described by Socransky (Socransky et al., 1998); none of these unclassified bacteria were detected in appreciable abundance across patients and sites. No statistically significant associations between IACSM type and specific bacteria either in the material itself or in the sulcus were detected by two different statistical approaches (ANOVA and Indicator Species Analysis). More specific analyses were therefore restricted to the color complexes.

Bar graphs depicting the relative proportions of the microbial complexes in each patient per each sample and site can be found in Figure 3a,b. Overall, there appeared to be a preponderance of early colonizing bacteria (Green, Yellow, Blue, and Purple complexes as described by Socranksy) within the IACSM, while the peri-implant sulcus appeared to be dominated by Socranksy's Orange complex bacteria (Socransky et al., 1998). Paired *t* tests were performed with a $p \leq .05$ denoting significant differences between IACSM and peri-implant sulci complex bacteria per site. The proportion of sequences from Red and Orange complex bacteria found on the IACSM was significantly less than in the peri-implant sulci ($p < .05$). The proportion of sequences from Green, Yellow, and Blue complex bacteria found on the occluding materials was significantly greater than in the peri-implant sulci ($p < .05$). There was no significant difference for the Purple complex. For sulcus samples, the proportion of sequences from Red, Purple, Yellow, and Blue complex bacteria was significantly different between patients ($p < .05$). The proportion of sequences from Orange and Green complex bacteria was not statistically different between patients ($p < .05$).

4 | DISCUSSION

This is the first clinical trial reporting the effects of different implant prosthesis access channel sealing materials (IACSM) on peri-implant parameters and on the bacterial flora of the screw channel and the peri-implant sulci. No implant and abutment system has been proven to be able to provide a completely sealed interface (Tettamanti et al., 2017). Having a microgap and microleakage at this interface facilitates plaque retention, possibly allowing the interface to act as a nidus of continual bacterial leakage and communication between the sulcus and the screw channel. This study supports the presence of microleakage at the level of the implant abutment interface and/or at the coronal seal of the prosthesis as meticulous protocols were followed to decontaminate the implants and prostheses, and to place and collect the samples from the patients in an aseptic manner. The methods used here resulted in no microbial growth from sterile materials, and it is likely that OTU or ASV identified from processed sterile paper points and unused dental materials were from DNA extracted from non-viable bacteria or reflected minor contamination

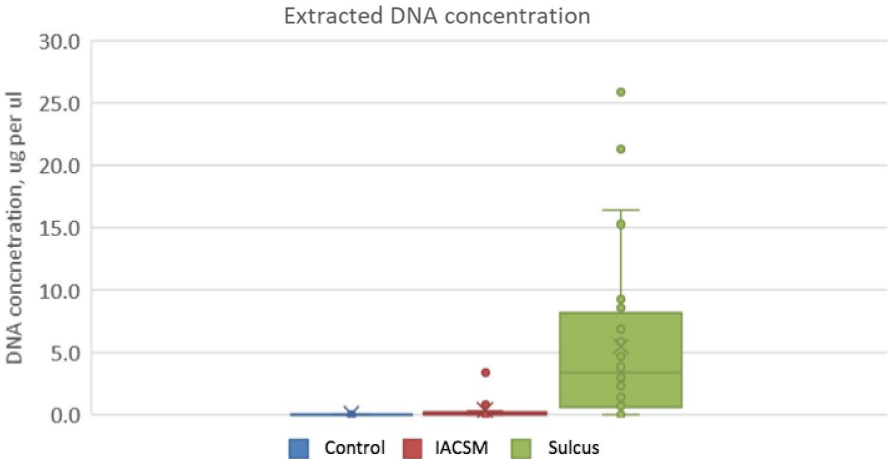


FIGURE 1 DNA concentrations determined from the control (sterile paper points and sterile IACSM), the implant abutment channel sealing materials (IACSM), and the peri-implant sulcus (paper points placed in the sulcus). Note that the concentrations for the control and IACSM samples were below the detection limit of the assay

of reagents combined with the high sensitivity of PCR, as the very general primers used here are able to amplify most known bacteria (and many archaea). In our study, the majority of prosthetic assemblies were screwed directly to the implant platform and all had butt-joint implant abutment interfaces. Minor differences in the exact interfaces for two out of the eight patients did not have statistically significant effects.

The data presented here suggest that there exist two dissimilar microbial composition at the screw channel/IACSM and the peri-implant sulci with some degree of communication. Early colonizer

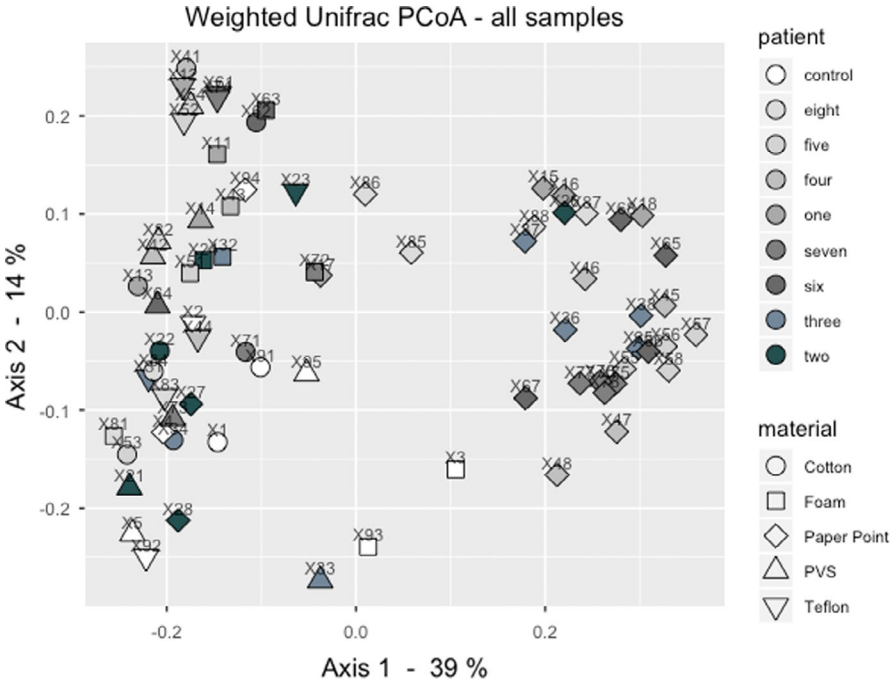


FIGURE 2 The first two axes of the Weighted UniFrac amplicon sequence variants (ASV) ordination captured 53% of the variance. The remaining axes (not shown) each captured less than 8% of the variance

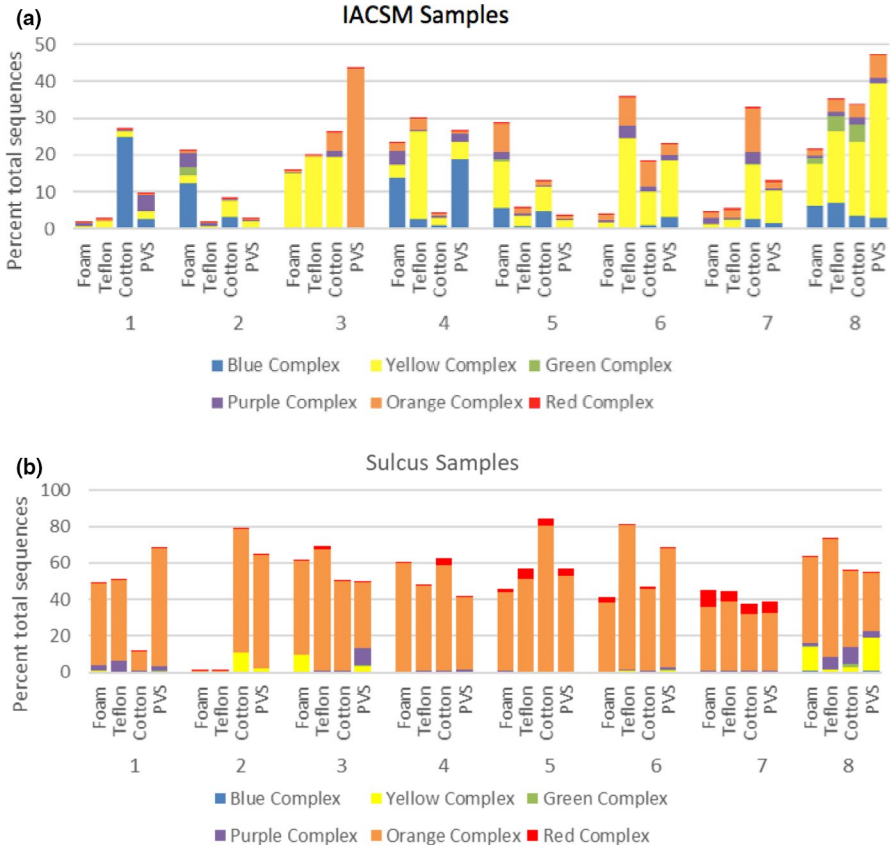


FIGURE 3 (a) Percentage of "Socransky" microbial complexes found on Implant Abutment Channel Sealing Materials (IACSM) samples from each of the 8 patients. (b) Percentage of "Socransky" microbial complexes found in the sulcus paper point samples from each of the 8 patients

type of bacterial species belonging to the Green, Yellow, Blue, and Purple complexes were mainly found on the IACSM while more members of the Orange and Red complexes were found in the peri-implant sulci (Socransky et al., 1998). It is possible that there is environmental selection for certain bacterial types to thrive in each location. Bacterial flora can be impacted by degree of inflammation, plaque abundance, time, pH, oxygen levels, and other factors (Takahashi, 2005). Kroger et al. demonstrated that increasing peri-implant sulcus depth is associated with dysbiosis of the biofilm and peri-implant disease (Kröger et al., 2018). While the IACSM and sulcus microbiomes were distinct (Figure 3a,b), the differences were mainly in relative abundances and there was substantial shared membership between them. Additionally, the amount of DNA extracted suggests that IACSM samples usually contained very low bacterial biomass compared to the sulcus samples. Based on this information, it is reasonable to assume that microleakage allows movement of bacteria from the peri-implant sulcus to the IACSM and/or vice versa. The greater quantity of the Orange and Red complex suggests that the biofilm at the peri-implant sulcus is more developed as the Red complex appears later in the development of the biofilm (Carranza et al., 2006).

No effect of different IACSM on the peri-implant sulcus microbiome was detected. This is not in accordance with previously published data suggesting that material type could have an effect on microleakage and therefore influence the microbiota in the peri-implant sulcus (Park et al., 2012). The major difference between the present study and those already published is that this is an in vivo clinical study utilizing each patient's unique microbiomes found in oral fluids as the passage fluid versus an in vitro model utilizing fuchsin dye as the passage fluid in many reported studies (Gross et al., 1999; da Silva-Neto et al., 2017). The presence of bacteria in the abutment screw channel can be affected by several factors, including passage, survival, and growth, whereas the fuchsin dye must only have a gap in which to flow.

Sequences from diverse species not included in the Socransky and Haffajee microbial complexes were a substantial component of the total recovered in this study. One of these "other" bacteria, *Atopobium* species, was the second most commonly detected taxon after *F. nucleatum*. *Atopobium* species are gram-positive facultative anaerobic bacteria, which are either rod-shaped or elliptical coccobacilli. *Atopobium* species have been described in the peri-implant literature (da Silva et al., 2014). They have also been detected in other pathologic conditions, such as bacterial vaginosis and endodontic lesions (Datcu, 2014; Hsiao et al., 2012).

5 | CONCLUSION

Within the limitation of this 6-month randomized clinical trial, the implant abutment channel sealing material had no significant difference on the microbial composition within the sealing materials or within the peri-implant sulcus. It was found that the variation of bacterial species was most dependent on the individual patient examined. In addition, no significant changes were found in the peri-implant parameters such as gingival index, plaque index, and probing depths during the study period between the different implant abutment channel sealing material groups. However, evidence of direct bacterial microleakage at the A-I was found and supports further clinical trials in this area of implant dentistry.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

AUTHOR CONTRIBUTIONS

Conceptualization (Brandon Katz, Caroline Rubino); Data curation (Caroline V. Rubino, Brandon G. Katz, Kylie Langlois, Stephen G. Walker, Jackie L. Collier); Formal analysis (Caroline V. Rubino, Brandon G. Katz, Kylie Langlois, Stephen G. Walker, Jackie L. Collier); Investigation (Caroline V. Rubino, Brandon G. Katz, Kylie Langlois, Stephen G. Walker, Jackie L. Collier); Methodology (Caroline V. Rubino, Brandon G. Katz, Kylie Langlois, Stephen G. Walker, Jackie L. Collier); Project administration (Stephen G. Walker, Jackie L. Collier, Howard H. Wang, Julio A. Carrion, Vincent J. Iacono, Srinivas R. Myneni); Resources (Stephen G. Walker, Jackie L. Collier, Howard H. Wang, Julio A. Carrion, Vincent J. Iacono, Srinivas R. Myneni); Supervision (Stephen G. Walker, Jackie L. Collier, Howard H. Wang, Julio A. Carrion, Vincent J. Iacono, Srinivas R. Myneni); Validation (Howard H. Wang, Julio A. Carrion, Vincent J. Iacono, Srinivas R. Myneni); Writing-original draft (Howard H. Wang, Srinivas R. Myneni); Writing-review & editing (Kylie Langlois, Stephen G. Walker, Jackie L. Collier, Howard H. Wang, Julio A. Carrion, Vincent J. Iacono, Srinivas R. Myneni). All authors approved the final version of the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in NCBI Sequence Read Archive (https://www.ncbi.nlm.nih.gov/Traces/study/) under BioProject accession number PRJNA649980.

ORCID

Kylie Langlois https://orcid.org/0000-0002-1541-2919
Srinivas R. Myneni https://orcid.org/0000-0002-5531-8206

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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